

MATH 4790: Quiz 2

February 6, 2026

Name: _____

Score: _____/15

Instructions: Show all work and justify your answers. Only calculators approved for Exam FM are permitted. Credit will not be given to just guessing the correct answer.

Question 1.

Emma deposits \$100 into an account that earns interest at a nominal rate of 12% compounded quarterly. Lucy deposits \$200 into an account that earns a constant force of interest of 6%. At time n , both accounts have the same amount. Find the amount both accounts have at time n .

(A) 275.5

(B) 300.5

(C) 319.5

(D) 408.5

(E) 467.5

Question 2.

Two accounts accumulate interest as follows.

Account 1: A deposit of \$200 is made at time $t = 0$ and the account earns simple interest at 6%.

Account 2: A deposit of \$100 is made at time $t = 0$ and the amount function is

$$A(t) = 100 + kt^2, \quad t \geq 0.$$

It is given that the amount of interest earned from time $t = 1$ to time $t = 3$ is the same for both accounts. Find k .

- (A) 3 (B) 4 (C) 5 (D) 6 (E) 7

Question 3.

Tony deposits \$50 into an account at time $t = 0$.

- During the first year, the account earns interest at an effective discount rate of 5%.
- During the next n years, the force of interest is given by

$$\delta_t = \frac{2}{4 + 2t}.$$

- During the final year, the account earns interest at an effective interest rate of 5%.

If the final value of the account is \$100, find n .

- (A) 1.92 (B) 2.43 (C) 3.05 (D) 3.51 (E) 5.17